

Can Public Artifacts Substantiate Financial-Agent Alpha? A 98-Work Evidence Audit and Descriptive Spanning Exercise

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Abstract

Most reported alpha claims for LLM financial agents do not hold up to independent scrutiny. A cutoff-bounded systematic screen yields 103 candidate lineages backed by 98 distinct cited works; frozen rules retain 69 formula-discovery or trading works and exclude 29 visible benchmarks, comparators, audits, or adjacent works. Forty retained works receive 50 good-faith common-task mappings; 29 remain availability-only. A separate 14-attempt code audit yields no native-agent replication and one testable released seed-expression adaptation, which does not beat a six-factor benchmark. Among 13 source-grounded component tests from five works, median annualized alpha is 0.88%, one estimate is nominally positive, and none survives within-subset Holm correction. The other 49 mappings are favorable narrative translations. At 10-basis-point one-way costs, family median alpha is -0.05% versus 2.15% gross; the sole conditional Holm result comes from a researcher-authored proxy, is specification-fragile, and has a simultaneous lower bound below two percentage points. A post-hoc 133-factor diagnostic yields no familywise result. The literature therefore substantially overstates the evidence for LLM-generated alpha: native agents are rarely reproducible, and the testable evidence does not demonstrate robust, economically material alpha. Because mappings were frozen after U.S. outcomes were inspected and had no independent second coder, these statistics are descriptive conditional diagnostics, not confirmatory inference. This conclusion concerns the reported evidence, not whether every unavailable agent must fail.

CCS Concepts

• **Computing methodologies** → **Multi-agent systems**; • **Applied computing** → **Economics**; • **General and reference** → *Empirical studies*.

Keywords

financial agents, alpha, asset pricing, reproducibility, multiple testing, empirical audit

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1 Introduction

LLM-agent papers increasingly claim that their systems can discover profitable factors, outperform benchmarks, or generate investment alpha [14, 44, 70, 75, 101]. Our audit finds that these claims generally do not withstand independent scrutiny. None of 14 targeted code audits reproduces a native agent, the closest source-grounded tests yield no familywise-adjusted evidence of alpha, and the few positive findings depend on researcher-authored proxies or fragile specifications.

Reported alpha can fail scrutiny in two ways. First, the system may be unevaluable because its code, data adapter, trained state, or dated output is unavailable. In that case, the alpha claim lacks reproducible support; we do not impute a zero return. Second, a testable return path may exist but fail an asset-pricing benchmark. We therefore use the highest-fidelity feasible route and label the estimand at every step. Native or code-backed returns test released implementations. When they are unavailable, reconstruction preserves source-supported content, chooses a favorable reasonable implementation, encodes ambiguity as variants, and discloses every substitution.

The target of inference is the reported evidence, not the unknowable performance of an unavailable agent. We ask whether public artifacts establish alpha under an independently specified common task. Our U.S. portfolios are retrospective spanning diagnostics conditional on researcher-authored mappings; they neither demonstrate prospective discovery nor correct for prior factor mining or publication selection [22]. This boundary does not weaken the conclusion: the literature presents alpha as an empirical result, and the released evidence generally fails to substantiate that result.

We make four contributions: a cutoff-bounded 98-work evidence waterfall; a direct-code audit separating one testable expression adaptation from non-evaluable attempts; a good-faith reconstruction contract with an explicit attribution boundary; and a 62-row U.S. ledger with cost, missing-return, spanning, and multiplicity diagnostics. The central finding is simple: most reported alpha claims do not hold up. The agents are rarely reproducible, the closest source-grounded evidence has no adjusted alpha, and the broader diagnostic's lone adjusted result is a fragile researcher-authored proxy rather than agent output.

2 Evidence Design

2.1 Denominators and inferential targets

The screen covers database inception through 2 August 2026. Three query families combine LLM/agent terms with alpha or strategy discovery; trading, portfolios, returns, or backtests; and benchmark, audit, artifact, or leakage terms. Sources are arXiv's relevant CS and q-fin categories, ACL Anthology, OpenReview, ACM DL, and SSRN, plus citation chasing from AlphaBench, AlphaQT-Bench, FINSABER, and the Agentic Trading survey. Crossref/OpenAlex resolve duplicates; GitHub locates only record-linked artifacts. The artifact releases queries, routes, rules, canonical records, deduplication notes, and row rationales. Raw hit counts, rankings, and timestamps were not preserved, so this is a cutoff-bounded systematic screen rather than a complete universe.

The registry has 103 lineages, 104 record URLs, and 98 works after six duplicate manifestations are collapsed. All 98 works are cited. Formula discovery (F) and trading agents (T) are retained; benchmarks/audits (B), community software (C), and mechanistic comparators (M) remain visible with reasons. Frozen rules exclude

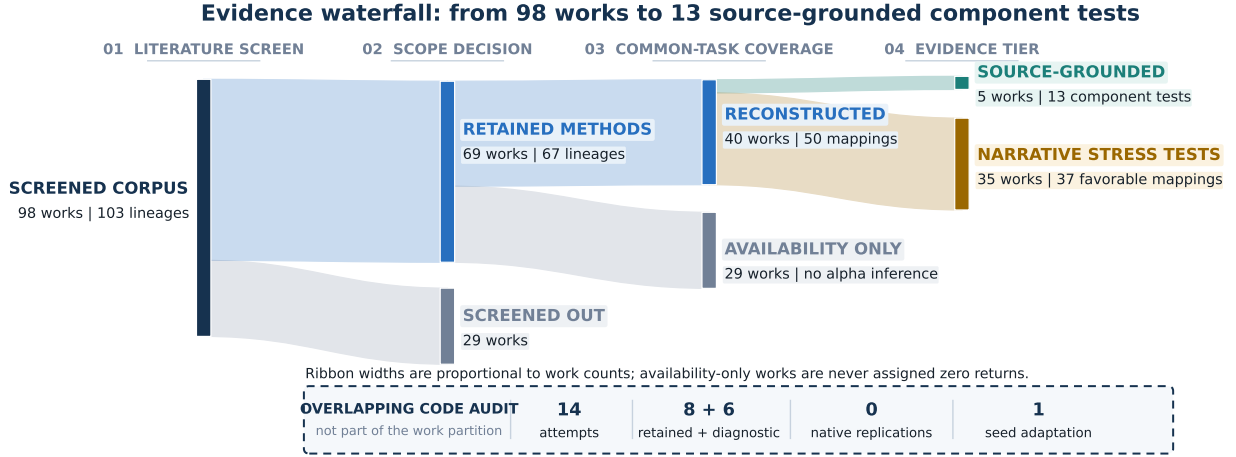


Figure 1: Work-level corpus and evidence waterfall. Ribbon widths are proportional to work counts as 98 canonical works narrow to 69 retained, 40 reconstructed, and five supporting 13 source-grounded component tests. The dashed-outline band reports the separate, overlapping code audit. A missing native path is an evidence failure, not a zero return, and a mechanism mapping tests historical economic content rather than agent performance.

29 works [10, 64, 65, 86, 96, 102] [8, 12, 13, 33, 35, 56, 88, 93] [19, 28, 32, 37, 51, 54, 58, 61, 83, 87, 91, 104, 107, 108, 110]. The 69 retained works in 67 lineages comprise 30 formula works [4, 9, 14, 27, 30, 44, 49, 53, 63, 67, 70, 73, 75, 78, 90, 99] [3, 21, 31, 42, 45, 48, 66, 71, 77, 79, 92, 95, 100, 101] and 39 trading works [15, 43, 74, 82, 97, 98, 103] [1, 2, 7, 17, 18, 29, 39–41, 46, 47, 50, 55, 60, 68, 69, 72, 80, 81, 84, 85, 106] [5, 6, 11, 16, 24, 26, 36, 38, 62, 76]. Thus 98/69/67 are screened works/retained works/lineages, not replication counts.

We construct 50 mappings for 40 retained works: five works support 13 source-grounded component tests and 35 support 37 narrative stress tests. The other 29 remain availability-only because no frozen, supported common-panel mapping exists; no proxy or zero alpha is imputed. Twelve mappings from 11 excluded diagnostic sources produce the prespecified 62-portfolio family. Its legacy discovery list has 55 records and 51 mapped indices (one names two systems). None states our exact monthly U.S. top-1,000 six-factor claim, so this is a common-task diagnostic, not 51 claim replications.

The overlapping code route covers eight retained works and six excluded diagnostics (Table 2). We complete zero native-agent replications. One QuantEvolver seed expression yields a testable code-backed adaptation; 13 blocked attempts never enter a performance denominator. Repository, license, execution, and performance status remain separate fields.

2.2 Artifact and direct-code route

The first reason reported alpha claims do not hold up is a reproducibility gap. We freeze URLs/revisions and require dated output from the released system for a native path. Of 67 retained lineages, 20 list an artifact, 18 are reachable, nine have a reuse license, and three ship dated native output; none matches the common task. Among 14 targeted implementations, one released QuantEvolver seed expression yields a testable code-backed adaptation, not a trained-agent run. Its annualized six-factor alpha is 0.33% ($t = 0.65$).

Table 1: Evidence hierarchy and what each result identifies.

Stage	Evidence/result	Permitted interpretation
Corpus screen	98 works → 69 retained + 29 excluded	All records cited; exclusions remain auditable
Retained coverage	69 works → 40 reconstructed + 29 availability-only	No proxy or zero return imputed to availability-only works
Targeted code audit	8 retained + 6 diagnostic attempts; 0 native; 1 adaptation	Expression-level result only; no performance claim for 13
Mechanism mapping	50 retained + 12 diagnostic mappings; 13 grounded, 49 narrative	Component evidence only; no narrative null charged to a source
Broad diagnostic	133 factors; 0 FWE results	Post-hoc diagnostic, not confirmatory evidence
International forensic audit	40 events; all dominated by one short-held return	Data/implementation alarm; no performance inference

The other 13 have explicit output, input/adaptor, task, or executability blockers. Thus no native-agent result is reproduced, and only one non-native adaptation enters the performance denominator.

2.3 Mechanism mapping and disclosed discretion

When direct evaluation is unavailable, we map public descriptions to point-in-time JKP characteristics [25]. This formalizes a good-faith, in-spirit reconstruction: preserve the strongest supported content, disclose every substitution, and give the idea the benefit of the doubt rather than construct an easy-to-reject surrogate.

Table 2: The 14 targeted implementation attempts. Verdicts assess whether released artifacts provide a testable native-agent return path; eight systems belong to the 67-system F/T census.

System	Released object / native task	Audit verdict and basis
AlphaAgent [70]	China A-share factor-mining code	Fail: no shipped dated strategy return series.
AlphaBench [52]	Formula-mining benchmark code	Fail: executable benchmark components, but no dated strategy returns.
AlphaForgeBench [105]	End-to-end strategy-design benchmark	Fail: aggregate metrics only; no dated common-task return stream.
AlphaPROBE [20]	Graph-based factor-mining comparator	Fail: no approved JKP adapter or released dated expressions/returns.
ContestTrade [69]	Sequential multi-agent trading code	Fail: no replay adapter or shipped common-task returns.
DeepFund [34]	Live fund-investment benchmark	Fail: no approved historical reproduction path for the common panel.
FinCon [98]	Sequential multi-agent trading system	Fail: available implementation was unofficial/ambiguous and task-incompatible.
GuruAgents [29]	Nasdaq-100 guru-persona portfolios	Fail: shipped non-JKP backtest is inspectable but incompatible with the common task.
LiveTradeBench [94]	Live stock/prediction-market benchmark	Fail: no frozen historical return path compatible with the evaluator.
QuantAgent [84]	Hourly/4-hour HFT across stocks, futures, and crypto	Fail: intraday native task cannot be represented by monthly JKP data.
QuantEvolver [101]	Factor-discovery framework plus seed expressions	Partial: one released seed expression is testable; $\alpha = 0.33\%$, $t = 0.65$. It is not a trained-agent run.
R&D-Agent-Quant [90]	General R&D-Agent framework and quant workflow	Fail: no shipped U.S. candidate returns or completed JKP adapter.
TradeTrap [89]	Trading-agent reliability harness	Fail: shipped results are a non-JKP diagnostic, not a common-task return path.
Trading-R1 [81]	RL-trained trading-reasoning system	Fail: public repository snapshot is a placeholder without executable system code.

The eight-step contract records a claim card; prioritizes released expressions, equations, examples/rules, narratives, then summaries; preserves supported ingredients, signs, lags, interactions, and constraints; and resolves ambiguity favorably without return tuning. Source-stated direction, explicit long-only or sparse top- k rules, and both gross and net results are preserved. Reasonable alternatives should be coded before outcomes and remain in the multiplicity family; no sign, candidate, or variant may be selected using test returns.

A source-protection rule lets released expressions, named rules, and examples speak only to that component under the common task. Narrative translations are favorable stress tests whose failure cannot count as evidence against the source. Preferred review uses a second, outcome-blind coder. The public card records supported and supplied elements, favorable choices, omissions, alternatives, freeze/outcome/reviewer status, and the negative-evidence boundary.

The mappings predate this contract, so we apply only its source-protection rule retrospectively. Each row records the source, rationale, score, portfolio rule, support, omissions, favorable choices, and permitted interpretation. Table 4 labels narrative translations M0 even when favorable.

The second reason the claims do not hold up is that the closest testable content shows no robust alpha. Combining M1 and M2 yields 13 source-grounded component tests. Their median six-factor alpha is 0.88%; one is nominally positive and none survives Holm within the 13-test subset. Under the post-hoc broad benchmark, median alpha is -2.00% , with zero nominal or Holm-positive results. The 49 M0 rows do not rescue the claim: their median alpha is -0.10% , and the sole full-family Holm result is a researcher-authored translation rather than agent output. Component tests do not reproduce native agents, but they show that the strongest reconstructable content does not deliver robust alpha under the common task.

The ledger makes three adverse facts explicit. First, 49 mappings are narrative translations and therefore barred from source-level negative inference; 12 have partial support and only one uses a released expression. Second, the registry was frozen after U.S. outcomes had been inspected, so its immutable hash does not make coding outcome-blind. Third, no independent second coder was

used. The mapping choice set is unknown; Holm and bootstrap calculations therefore quantify sampling variation only conditional on these 62 realized formulas. We treat every resulting p -value and interval as descriptive, not confirmatory, even when a conditional adjustment is below 5%.

We can measure only a lower bound on that discretion. Five sources already have multiple coded mappings, covering 15 candidates. Selecting one mapping per source yields 144 combinations; the same AlphaAgent-inspired quality proxy is the sole Holm discovery in every combination. This does not validate singly mapped formulas, nor does it replace independent coding. It shows only that the headline count is unchanged within the alternatives that happened to be coded.

2.4 Common U.S. task and descriptive estimators

The primary panel is monthly U.S. common equity from August 2001 through December 2024. At formation month t , the universe $\mathcal{U}_t$ contains the 1,000 largest positive-market-equity securities, and mapping i produces a point-in-time score $s_{ijt} = g_i(x_{jt})$. For the usual value-weighted long-short mapping, let H_{it} and L_{it} denote its top and bottom deciles and let $m_{jt} > 0$ be market equity. The frozen target weights are

$$w_{ijt} = \begin{cases} m_{jt} / \sum_{k \in H_{it}} m_{kt}, & j \in H_{it}, \\ -m_{jt} / \sum_{k \in L_{it}} m_{kt}, & j \in L_{it}, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

Thus each leg carries unit notional; labeled long-only and sparse top- k mappings use their stated rules. Formation never conditions on next-month return availability. Define $R_{j,t+1}^{e,*} = R_{j,t+1}^e$ when the held excess return is observed and $R_{j,t+1}^{e,*} = 0$ otherwise; weights are never renormalized using future coverage. A severe stress instead assigns $-\text{sgn}(w_{ijt})$ to a missing return, or -100% for a long and $+100\%$ for a short; it is not a delisting-return estimate.

Table 3: The five papers underlying the 13 source-grounded component tests. “Source-grounded” applies to the named component, not the full native agent or original evaluation.

Source (N mappings)	Original idea and task	Component implemented in our common task
EFS [53] (3)	LLM evolutionary factor search for sparse top- m portfolios on Fama–French and US50/HSI45/CSI300 datasets.	Momentum/low-volatility/breakout score; reversal/low-noise score; and a sparse equal-weight top-five variant.
QuantEvolver [101] (1)	Reinforcement fine-tuning generates executable factor-DSL expressions across market benchmarks.	The released risk-adjusted-momentum seed $\text{ret}_{12,1}/ \text{rvol}_{252} $; portfolio formation remains ours.
Alpha-Jungle [67] (2)	LLM-guided MCTS searches symbolic stock factors using price, volume, and volatility operators.	A price/volume/momentum score and a trend-after-volatility-compression score based on documented examples.
FAMA [44] (1)	Neural-symbolic factor mining evaluated by RankIC/RankICIR and an S&P 500 investment simulation.	A simple interpretable value–momentum–profitability score motivated by the paper’s factor examples.
GuruAgents [29] (6)	Prompt-guided agents emulate five named investor philosophies in a Nasdaq-100 backtest.	Graham, Buffett, Greenblatt, Piotroski, and Altman characteristic screens plus their equal-weight ensemble.

Table 4: Mapping fidelity and U.S. results by tier.

Evidence tier	N	Median alpha	Cond. Holm +
M0 narrative translation	49	−0.10%	1
M1 example/motif support	6	−0.71%	0
M1 named-rule support	6	3.12%	0
M2 released seed expression	1	−0.28%	0

Annualized six-factor alpha net of 10 bp one-way costs. Holm values are conditional on the realized mappings and are descriptive.

Turnover restores target exposure from drifted pre-trade weights. With total security return $R_{j,t}^{\text{tot}}$,

$$\begin{aligned}
\tilde{w}_{ij,t} &= \frac{w_{ij,t-1}(1 + R_{j,t}^{\text{tot}})}{1 + \sum_k w_{ik,t-1}R_{k,t}^{\text{tot}}}, \\
\tau_{it} &= \sum_j |w_{ijt} - \tilde{w}_{ij,t-}|, \\
r_{i,t+1}^{\text{net}}(c) &= \sum_j w_{ijt}R_{j,t+1}^{e,*} - c\tau_{it}.
\end{aligned} \tag{2}$$

The one-way rate c spans $\{0, 5, 10, 25, 50\} \times 10^{-4}$, with 10 bp primary. This linear model omits borrow, financing, locate constraints, nonlinear impact, and capacity. Benchmark factors remain gross, so costed-candidate alpha is a deliberately conservative asymmetric comparison.

For each executable candidate, we estimate the six-factor model with $f_t \in \mathbb{R}^6$ and its HAC statistic

$$\begin{aligned}
r_{i,t}^{\text{net}} &= \alpha_i + \beta_i^\top f_t + \varepsilon_{i,t}, \\
T_i &= \frac{\hat{\alpha}_i}{\widehat{\text{se}}_{\text{NW}}(\hat{\alpha}_i)}, \quad L_T = \lfloor 4(T/100)^{2/9} \rfloor.
\end{aligned} \tag{3}$$

The Newey–West rule gives $L_T = 4$ at $T = 281$ [59]. For ordered two-sided p -values $p_{(1)} \leq \dots \leq p_{(m)}$, Holm adjustment is

$$p_{(k)}^{\text{H}} = \max_{\ell \leq k} \min\{1, (m - \ell + 1)p_{(\ell)}\}, \quad m = 62, \tag{4}$$

where non-estimable planned paths enter with $p = 1$ [23]. Paired moving-block draws jointly resample candidate and factor months. For $B = 2,000$, block length six, and fixed original HAC standard

errors, the single-step family statistic is

$$\begin{aligned}
T_i^{*(b)} &= \frac{\hat{\alpha}_i^{*(b)} - \hat{\alpha}_i}{\widehat{\text{se}}_{\text{NW}}(\hat{\alpha}_i)}, & M^{*(b)} &= \max_i |T_i^{*(b)}|, \\
\hat{p}_i^{\max} &= \frac{1 + \sum_{b=1}^B \mathbf{1}\{M^{*(b)} \geq |T_i|\}}{B + 1}, & \text{LB}_i^{\text{sim}} &= 12 (\hat{\alpha}_i - q_{.95} \widehat{\text{se}}_{\text{NW},i}).
\end{aligned} \tag{5}$$

Here $q_{.95}$ is the 95th percentile of M^* . Block lengths three and twelve and HAC lags 0, 3, 6, and 12 are sensitivities. A simultaneous lower bound of at least two annual percentage points is a pragmatic reporting threshold selected after U.S. outcomes were available, not a preregistered rule.

3 Results

3.1 Reported alpha under the common U.S. task

The common-task results do not support the reported alpha. Among the 13 source-grounded component tests, median annualized six-factor alpha is 0.88%; one estimate is nominally positive and none survives within-subset Holm correction. Expanding to all 62 executable paths does not change the conclusion. At 10 bp, median alpha is −0.05% (interquartile range [−1.26%, 2.05%]); 30 estimates are positive, 6 have nominal $p < .05$, and one has conditional Holm and maximum- $|t|$ values below .05. No simultaneous lower bound exceeds two percentage points. The sole adjusted result is the largest estimate, 6.93%, from a *researcher-authored AlphaAgent-inspired quality proxy*; the software did not generate it, and its simultaneous lower bound is only 0.48%.

That lone result is statistically fragile. Across fixed Newey–West lags 0/3/6/12, nominal positive counts are 5/6/6/6 and Holm counts are 0/1/1/0: the adjusted count is one only at lags 3 and 6. Block-length changes leave the maximum-statistic count at one, but they do not address mapping discretion or publication selection. A result that depends on a researcher-authored proxy and selected serial-correlation specifications is not robust evidence of agent-generated alpha.

Trading costs further weaken the claim. Median alpha falls from 2.15% gross to 1.01% at 5 bp and −0.05% at 10 bp; positive counts across 0/5/10/25/50 bp are 46/42/30/18/10, while conditional Holm counts are 1/1/1/0/0. Among 46 gross-positive mappings, median break-even cost is 21.4 bp (IQR 8.2–42.1). Even gross returns produce only one conditional adjusted result, and no simultaneous lower bound reaches the two-percentage-point materiality threshold.

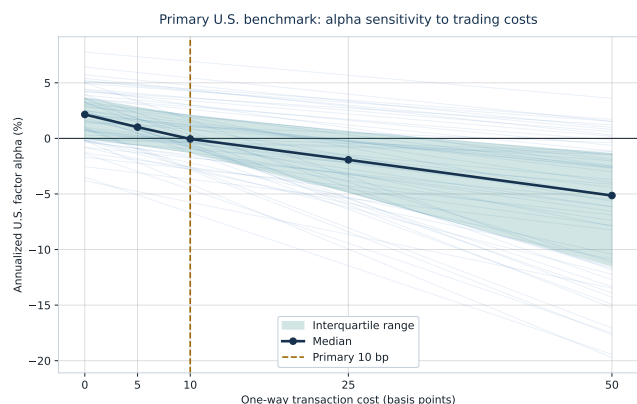


Figure 2: Annualized six-factor alpha across one-way trading-cost assumptions. Thin lines are candidate mappings; the band is the interquartile range. Benchmark factors remain gross.

Missing-return exposure is small in weight but economically nontrivial under an extreme imputation. The median mapping’s mean missing gross-weight share is 0.20% (maximum across mappings 0.43%). The position-adverse unit-move stress changes family median alpha to -4.84%, leaves 4 positive estimates, and leaves zero nominal or conditional Holm positives. This does not identify delisting returns; it shows that zero fill is consequential and that adverse missing outcomes do not create evidence of alpha.

Taken together, the evidence explains why the reported alpha does not hold up. The closest source-grounded tests provide no familywise evidence of incremental return, and the broader family produces only one fragile result that cannot be attributed to an agent. These findings reject the strength of the published evidence; they do not assign zero returns to unavailable native systems.

3.2 Post-hoc broad-factor diagnostic

The broader factor benchmark does not rescue the alpha claim. It uses the market plus 132 JKP factors. The source panel ends in December 2021; after shifting factors to return months and a 120-month trailing window, evaluation spans August 2011–January 2022 (126 months), not 2024.

Each month selects the ridge penalty on 96 past fit and 24 validation months, refits on all 120, and emits a one-month-ahead residual. The 5,000-replication block bootstrap resamples already-estimated residuals; it does *not* rerun rolling estimation or penalty selection. Conditional on this possibly optimistic realized path and designed after primary outcomes, it is descriptive, not robustness confirmation.

There are 23 positive estimates and one nominal result, but zero conditional Holm or maximum- $|t|$ results and zero simultaneous lower bounds above two percentage points. The best raw p -value is 0.0025; its Holm value is 0.152 and maximum-statistic value is 0.091. This is directionally consistent with spanning, but remains a post-hoc diagnostic over a truncated period.

3.3 International extension: forensic result, not alpha evidence

The international extension is excluded from headline performance inference. Thirty-five paths cross nonpositive NAV, producing 40 events (31 France, five Canada, two Germany, one Britain, one Italy). Security-level reconstruction matches every event; in all 40, one extreme positive return held short causes at least half the loss. Two France month-cells contain 24 events.

Return caps of +100%/+200%/+500%/+1,000% leave 0/1/1/7 events, but are not admissible corrections. Corporate actions, return construction, stale prices, or short-weight interactions may be responsible. Without independent security-level validation, this is a data/implementation alarm; the ledger retains every event, identifier, weight, return, contribution, error, and cap diagnostic.

4 Why the Reported Alpha Claims Do Not Hold Up

Three findings drive the conclusion. First, public releases do not reproduce any native agent in 14 targeted audits. Second, the 13 closest source-grounded component tests produce no familywise-adjusted alpha. Third, the sole adjusted result in the broader family is a researcher-authored proxy, disappears under some HAC choices, and never clears the familywise materiality threshold. The literature therefore substantially overstates the evidence for LLM-generated alpha.

This conclusion is strong but specific: most reported alpha claims do not survive independent evidentiary scrutiny. It is not a claim that every unavailable native agent must lose money. Many systems postdate the December 2024 return panel, and mappings were frozen in 2026 after outcome inspection. The exercise cannot distinguish discovery from repackaging of known characteristics. Holm adjustment covers sampling multiplicity only across the 62 realized formulas; it does not cover researcher mapping choices, the literature’s search process, unpublished mappings, or publication selection. Without an independently coded, outcome-blind reconstruction, performance results remain descriptive.

Common-task comparability omits language, memory, retrieval, debate, native timing/execution, dynamic allocation, non-equities, and tool adaptation. A mapping failure cannot prove native-agent failure, and a mapping success cannot be attributed to an agent. This limitation matters for agent-level claims but does not rehabilitate the published evidence: authors who report alpha bear the burden of releasing artifacts that make the result independently testable.

Benchmark practice outside finance offers a model: ABC separates task/outcome validity and reporting [109]; Terminal-Bench uses containerized coding tasks, executable tests, human solutions, and failure analysis [57]. Financial-agent releases additionally need pinned code/environments; dated signals, positions, or returns; universe, timing, cost, financing, model, prompt, and seed definitions; search breadth and failures; and provenance or ablation separating agent contribution from factor libraries and human search spaces.

5 Reproducibility and Audit Trail

The ledger contains the search protocol/routes; 103 lineages, 104 URLs, and 98 dispositions; 14 code attempts; the five-paper/13-component inventory; 62 formulas/decisions; aggregate U.S. returns; cost, missing-return, broad, and international diagnostics; scripts, tests, and hashes. It states that raw hits were not preserved, mappings were not outcome-blind, and no second coder was used. Restricted security data are referenced by path and hash.

Deterministic builders regenerate macros/figures from verified outputs. Tests cover timing, weights, drift, costs, inference, mapping sensitivity, citations, and event reconstruction. Claims do not depend on an unsubmitted appendix.

Responsible use and AI assistance. No orders were transmitted. Results are retrospective research evidence, not investment advice. Generative AI assisted literature organization, code drafting, diagnostic review, and prose editing; the human author is responsible for source selection, mappings, protocol choices, statistical specifications, citation verification, and all reported claims.

6 Conclusion

Most reported alpha claims for LLM financial agents do not hold up to independent scrutiny. The 98-work screen narrows to 69 retained works, 40 reconstructed works, and five works supporting source-grounded component tests. The code route reproduces no native agent; its sole testable seed adaptation does not beat the benchmark. Among 13 source-grounded components, one estimate is nominally positive and none is Holm-positive. The broader 62-row family yields one conditional result, but it comes from a researcher-authored proxy, is HAC-fragile, and does not clear the materiality threshold. The post-hoc 133-factor diagnostic yields no adjusted result, while international performance is excluded after a plausibility failure.

The message is not that every LLM agent is incapable of producing alpha. It is that the literature's empirical case is much weaker than its claims: native systems are rarely reproducible, and the strongest testable evidence does not demonstrate robust, economically material alpha. The paper supplies an evidence hierarchy and an inspectable workflow for separating credible alpha evidence from unsupported performance claims.

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